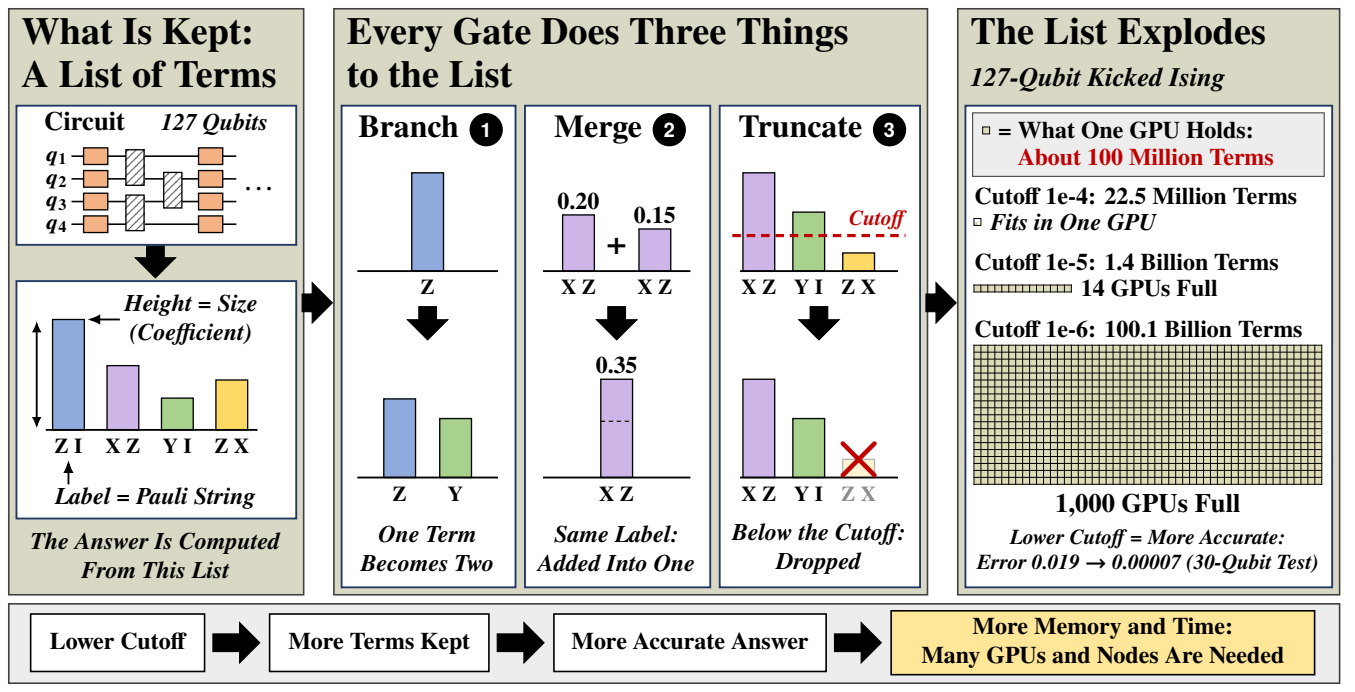

What Is Kept:
A List of Terms
Circuit
127 Qubits
q1
q2
q3
q4
Height = Size
(Coefficient)
Z I
X Z
Y I
Z X
Label = Pauli String
The Answer Is Computed
From This List
Every Gate Does Three Things
to the List
Branch 1
Z
Z
Y
One Term
Becomes Two
Merge 2
0.20
0.15
X Z
X Z
0.35
X Z
Same Label:
Added Into One
Truncate 3
Cutoff
X Z
Y I
Z X
X Z
Y I
Z X
Below the Cutoff:
Dropped
The List Explodes
127-Qubit Kicked Ising
= What One GPU Holds:
About 100 Million Terms
Cutoff 1e-4: 22.5 Million Terms
Fits in One GPU
Cutoff 1e-5: 1.4 Billion Terms
14 GPUs Full
Cutoff 1e-6: 100.1 Billion Terms
1,000 GPUs Full
Lower Cutoff = More Accurate:
Error 0.019 → 0.00007 (30-Qubit Test)
Lower Cutoff
More Terms Kept
More Accurate Answer
More Memory and Time:
Many GPUs and Nodes Are Needed